Chapter 3, Section 4

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July 20, 2026

1. Let $f : X \to Y$ be an affine morphism of Noetherian separated schemes (II, Ex. 5.17). Show that for any quasi-coherent sheaf $\mathcal{F}$ on X, there are natural isomorphisms for all $i \geqslant 0$,

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F}).$$

Proof. Let $\mathfrak{V}$ be an open affine cover of Y so that $\mathfrak{U} = (f^{-1}(V))_{V \in \mathfrak{V}}$ is an open affine cover of X. We can compute $H^i(X, \mathcal{F})$ using the Čech complex defined by the open covering $\mathfrak{U}$. Also, $f_*\mathcal{F}$ is a quasi-coherent $\mathcal{O}_Y$-module by (II, 5.8), so the cohomology of $f_*\mathcal{F}$ can be computed via the Čech complex defined by $\mathfrak{V}$ (4.5). Lastly, $f_*\mathcal{F}(V) = \mathcal{F}(U)$ for all $V \in \mathfrak{V}$ and corresponding $U = f^{-1}(V) \in \mathfrak{U}$, so the Čech cohomology of $\mathcal{F}$ and $f_*\mathcal{F}$ with respect to $\mathfrak{U}$ and $\mathfrak{V}$, respectively, are isomorphic. Hence, $H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F})$ for all $i \geqslant 0$ by (4.5). □

2. Prove Chevalley's theorem: Let $f : X \to Y$ be a finite surjective morphism of Noetherian separated schemes, with X affine. Then Y is affine.

(a) Let $f : X \to Y$ be a finite surjective morphism of integral Noetherian schemes. Show that there is a coherent sheaf $\mathcal{M}$ on X, and a morphism of sheaves $\alpha : \mathcal{O}_Y^r \to f_*\mathcal{M}$ for some $r > 0$, such that α is an isomorphism at the generic point of Y.

(b) For any coherent sheaf $\mathcal{F}$ on Y, show that there is a coherent sheaf $\mathcal{G}$ on X, and a morphism $\beta : f_*\mathcal{G} \to \mathcal{F}^r$ which is an isomorphism at the generic point of Y.

(c) Now prove Chevalley's theorem.

Proof.

(a) Let $\mathcal{L}$ and $\mathcal{K}$ be the sheaf of total quotient rings of X and Y, respectively. Since the function field of X is a finite field extension of that of Y, there exists an isomorphism $\phi : \mathcal{K}^r \to f_*\mathcal{L}$ for some $r > 0$. Let $\alpha : \mathcal{O}_Y^r \to f_*\mathcal{L}$ be the compositon of the natural map $\mathcal{O}_Y^r \to \mathcal{K}^r$ with ϕ. This is an isomorphism at the generic point of Y. Replacing $\mathcal{L}$ by the sub-$\mathcal{O}_X$-module spanned by the image of $\mathcal{K}^r$, we obtain the desired $\mathcal{M}$.

(b) Let $\mathcal{F}$ be any coherent sheaf $\mathcal{F}$ on Y, $\mathcal{A} = f_*\mathcal{O}_X$, and $\alpha : \mathcal{O}_Y^r \to f_*\mathcal{M}$ as in (a) for some coherent sheaf $\mathcal{M}$ on X. Applying $\mathcal{H}om(\cdot, \mathcal{F})$ to α gives $\beta : \mathcal{H}om(f_*\mathcal{M}, \mathcal{F}) \to \mathcal{F}^r$, which is an isomorphism at the generic point of Y. The $\mathcal{O}_Y$-module $\mathcal{H}om(f_*\mathcal{M}, \mathcal{F})$ is naturally a coherent $\mathcal{A}$-module, and finite morphisms are also affine, so by the equivalence of categories of coherent $\mathcal{O}_X$-modules and coherent $\mathcal{A}$-modules (II, 5.17e), there exists a coherent sheaf $\mathcal{G}$ on X such that $f_*\mathcal{G} \cong \mathcal{H}om(f_*\mathcal{M}, \mathcal{F})$.

(c) Let X_{red} and Y_{red} be the reduced schemes of X and Y, respectively (II, Ex. 2.3). By the universal property of $Y_{\text{red}} \to Y$, there exists a unique morphism $X_{\text{red}} \to Y_{\text{red}}$ making the following diagram commute

$$\begin{tikzcd} X_{\text{red}} \arrow[r] \arrow[d] & Y_{\text{red}} \arrow[d] \\ X \arrow[r] & Y \end{tikzcd}$$

A scheme is affine if and only if its reduced scheme is affine (Ex. 3.1) so assume X and Y are reduced by replacing $X \to Y$ with $X_{\text{red}} \to Y_{\text{red}}$. Because f is surjective, every irreducible component of Y has a nonempty irreducible preimage and has a structure of a closed subscheme of X. All closed subschemes of X are affine (II, Ex. 3.11b), so we reduce to the case X and Y integral.

Let $\mathcal{F}$ be a coherent sheaf on Y. We will show $H^i(Y, \mathcal{F}^r) = 0$ for all $i > 0$ (2.9) (3.7). Let $\beta : f_*\mathcal{G} \to \mathcal{F}^r$ as in (b). Suppose the set Σ of proper closed subsets of Y which are not affine is nonempty, and let $Z \subset Y$ be a minimal element in Σ. Thus, all proper closed subsets of Z are affine. By replacing X and Y with $f^{-1}Z$ and Z, and $\mathcal{F}$ with any coherent sheaf $\mathcal{F}$ with support in Z, we show Z must be affine, i.e., assume every proper closed subset of Y is affine, and by noetherian induction, this implies Y itself is affine.

By (II, 5.7), $\ker\beta$ and $\operatorname{coker}\beta$ are coherent sheaves on Y, so $K = \operatorname{Supp}(\ker\beta)$ and $C = \operatorname{Supp}(\operatorname{coker}\beta)$ are closed subsets of Y. Also, K and C are both proper subsets because β is an isomorphism at the generic point, so by hypothesis, K and C are affine. For any closed subset $Z \subseteq Y$, let $j : Z \to Y$ denote the inclusion map. We have $\beta \cong j_*j^*\ker\beta$ and similarly for $\operatorname{coker}\beta$, so by (3.7) and (Ex. 4.1),

$$H^i(Y, \ker\beta) \cong H^i(K, j^*\ker\beta) = 0, \quad H^i(Y, \operatorname{coker}\beta) \cong H^i(C, j^*\operatorname{coker}\beta) = 0.$$

The exact sequence

$$0 \longrightarrow \ker\beta \longrightarrow f_*\mathcal{G} \longrightarrow \mathcal{F}^r \longrightarrow \operatorname{coker}\beta \longrightarrow 0$$

induces two long exact sequences in cohomology

$$0 \longrightarrow H^0(Y, \ker\beta) \longrightarrow H^0(Y, f_*\mathcal{G}) \longrightarrow H^0(Y, \operatorname{im}\beta) \longrightarrow \cdots,$$

$$0 \longrightarrow H^0(Y, \operatorname{im}\beta) \longrightarrow H^0(Y, \mathcal{F}^r) \longrightarrow H^0(Y, \operatorname{coker}\beta) \longrightarrow \cdots.$$

Again by (Ex. 4.1), $H^i(Y, f_*\mathcal{G}) \cong H^i(X, \mathcal{G}) = 0$. By exactness of above, we conclude that $H^i(Y, \mathcal{F}^r) = 0$.

□

3. Let $X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$, and let $U = X - \{(0, 0)\}$. Using a suitable cover of U by open affine subsets, show that $H^1(U, \mathcal{O}_U)$ is isomorphic to the k-vector space spanned by $\{x^i y^j \mid i, j < 0\}$. In particular, it is infinite-dimensional.

Proof. Let $\mathfrak{U}$ be the open covering by the two open sets $V = X - \{x = 0\}$ and $U = X - \{y = 0\}$, with affine coordinates obtained by restricting the ones from X. Then the Čech complex has only two terms:

$$C^0 = \Gamma(V, \mathcal{O}_V) \times \Gamma(W, \mathcal{O}_W),$$
$$C^1 = \Gamma(V \cap W, \mathcal{O}_{V \cap W}).$$

Now

$$\Gamma(V, \Omega) = k\left[x, \frac{1}{x}, y\right]$$
$$\Gamma(W, \Omega) = k\left[x, y, \frac{1}{y}\right]$$
$$\Gamma(V, W, \Omega) = k\left[x, y, \frac{1}{x}, \frac{1}{y}\right]$$

and the map $d : C^0 \to C^1$ is given by $(f, g) \mapsto f - g$. To compute H^1, the image of d is the set of all expressions $x^i y^j$ where at least one of i, j is non-negative. Hence, $H^1(U, \mathcal{O}_U)$ is spanned by $\{x^i y^j \mid i, j < 0\}$. □

4. On an arbitrary topological space X with an arbitrary Abelian sheaf $\mathcal{F}$, Čech cohomology may not give the same result as the derived functor cohomology. But here we show that for H^1, there is an isomorphism if one takes the limit over all coverings.

(a) Let $\mathfrak{U} = (U_i)_{i \in I}$ be an open covering of the topological space X. A *refinement* of $\mathfrak{U}$ is a covering $\mathfrak{V} = (V_j)_{j \in J}$, together with a map $\lambda : J \to I$ of the index sets, such that for each $j \in J$,

$V_j \subseteq U_{\lambda(j)}$. If $\mathfrak{V}$ is a refinement of $\mathfrak{U}$, show that there is a natural induced map on Čech cohomology for any Abelian sheaf $\mathcal{F}$, and for each i,

$$\lambda^i : \check{H}^i(\mathfrak{U},\mathcal{F}) \to \check{H}^i(\mathfrak{V},\mathcal{F}).$$

The coverings of X form a partially ordered set under refinement, so we can consider the Čech cohomology in the limit

$$\varinjlim_{\mathfrak{U}} \check{H}^i(\mathfrak{U},\mathcal{F}).$$

(b) For any Abelian sheaf $\mathcal{F}$ on X, show that the natural maps (4.4) for each covering

$$\check{H}^i(\mathfrak{U},\mathcal{F}) \to H^i(X,\mathcal{F})$$

are compatible with the refinement maps above.

(c) Now prove the following theorem. Let X be a topological space, $\mathcal{F}$ a sheaf of Abelian groups. Then the natural map

$$\varinjlim_{\mathfrak{U}} \check{H}^1(\mathfrak{U},\mathcal{F}) \to H^1(X,\mathcal{F})$$

is an isomorphism.

Proof.

(a) An element $\alpha \in C^p(\mathfrak{U},\mathcal{F})$ is determined by giving an element

$$\alpha_{i_0,\dots,i_p} \in \mathcal{F}(U_{i_0,\dots,i_p}),$$

for each $(p+1)$-tuple $i_0 < \cdots < i_p$ of elements of I. We define the map $\Lambda^p : C^p(\mathfrak{U}) \to C^p(\mathfrak{V})$ by setting

$$(\Lambda\alpha)_{j_0,\dots,j_p} = \alpha_{\lambda(j_0),\dots,\lambda(j_p)}|_{V_{j_0,\dots,j_p}}$$

if $\lambda(j_i)$ are all distinct, and 0 otherwise. One has $d_{\mathfrak{V}}\Lambda = \Lambda d_{\mathfrak{U}}.$, so Λ induces a map of cohomology groups.

(b) We want to show the following diagram commutes

$$\begin{tikzcd}
 & \mathcal{I}^\bullet & \\
\mathcal{C}^\bullet(\mathfrak{U},\mathcal{F}) \arrow[ur] \arrow[rr] & & \mathcal{C}^\bullet(\mathfrak{V},\mathcal{F}) \arrow[ul]
\end{tikzcd}$$

where $0 \to \mathcal{F} \to \mathcal{I}^\bullet$ is any injective resolution of $\mathcal{F}$ in $\mathfrak{Ab}(X)$. This follows from the universal property of $\mathcal{I}^\bullet$. In particular, there is only one extension of $\mathcal{F} \to \mathcal{I}^\bullet$ to $\mathcal{C}^\bullet(\mathfrak{U},\mathcal{F})$.

(c) Embed $\mathcal{F}$ in a flasque sheaf $\mathcal{G}$, and let $\mathcal{R} = \mathcal{G}/\mathcal{F}$, so that we have an exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{R} \longrightarrow 0.$$

Define a complex $D^\bullet$ by

$$0 \longrightarrow C^\bullet(\mathfrak{U},\mathcal{F}) \longrightarrow C^\bullet(\mathfrak{U},\mathcal{F}) \longrightarrow D^\bullet(\mathfrak{U}) \longrightarrow 0.$$

For any refinement $\mathfrak{V}$ of $\mathfrak{U}$, we have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & C^\bullet(\mathfrak{U},\mathcal{F}) \arrow[r] \arrow[d] & C^\bullet(\mathfrak{U},\mathcal{G}) \arrow[r] \arrow[d] & D^\bullet(\mathfrak{U}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & C^\bullet(\mathfrak{V},\mathcal{F}) \arrow[r] & C^\bullet(\mathfrak{V},\mathcal{G}) \arrow[r] & D^\bullet(\mathfrak{V}) \arrow[r] & 0
\end{tikzcd}$$

Let $H^i_D(\mathfrak{U})$ be the cohomology of the complex $D^\bullet(\mathfrak{U})$. The natural map of complex

$$D^\bullet(\mathfrak{U}) \to C^\bullet(\mathfrak{U},\mathcal{R})$$

induces a natural map of cohomology groups $H^i_D(\mathfrak{U}) \to \check{H}^i(\mathfrak{U}, \mathcal{R})$. Noting that the Čech cohomology of $\mathcal{G}$ vanishes for $i > 0$, consider the following commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \varinjlim_{\mathfrak{U}} \check{H}^0(\mathfrak{U}, \mathcal{F}) \arrow[r] \arrow[d, "\wr"] & \varinjlim_{\mathfrak{U}} \check{H}^0(\mathfrak{U}, \mathcal{G}) \arrow[r] \arrow[d, "\wr"] & \varinjlim_{\mathfrak{U}} H^0_D(\mathfrak{U}) \arrow[r] \arrow[d] & \varinjlim_{\mathfrak{U}} \check{H}^1(\mathfrak{U}, \mathcal{F}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & H^0(X, \mathcal{F}) \arrow[r] & H^0(X, \mathcal{G}) \arrow[r] & H^0(X, \mathcal{R}) \arrow[r] & H^1(X, \mathcal{F}) \arrow[r] & 0
\end{tikzcd}$$

By the Snake lemma, it suffices to show

$$\varinjlim_{\mathfrak{U}} H^0_D(\mathfrak{U}) \cong H^0(X, \mathcal{R}).$$

□

5. For any ringed space $(X, \mathcal{O}_X)$, let Pic X be the group of isomorphism classes of invertible sheaves (II, §6). Show that Pic $X \cong H^1(X, \mathcal{O}_X^*)$ where $\mathcal{O}_X^*$ denotes the sheaf whose sections over an open set U are the units in the ring $\Gamma(U, \mathcal{O}_X)$, with multiplication as the group operation.

Proof. For any invertible sheaf $\mathcal{L}$ on X, let $\mathfrak{U} = (U_i)_{i \in I}$ be an open cover of X such that $\mathcal{L}|_{U_i}$ is free for each i. Fix isomorphisms $\varphi_i : \mathcal{O}_{U_i} \to \mathcal{L}|_{U_i}$. Then on $U_i \cap U_j$, we get an isomorphism $\varphi_i^{-1} \circ \varphi_j$ of $\mathcal{O}_{U_i \cap U_j}$ with itself. In particular, $(\varphi_i^{-1} \circ \varphi_j)(1)$ is a unit in $\mathcal{O}_{U_i \cap U_j}$. For another $U_k \in \mathfrak{U}$, one has

$$(\varphi_j^{-1} \circ \varphi_k)(1)\Big|_{U_i \cap U_j \cap U_k} \cdot (\varphi_i^{-1} \circ \varphi_j)(1)\big|_{U_i \cap U_j \cap U_k} = (\varphi_i^{-1} \circ \varphi_k)(1)\big|_{U_i \cap U_j \cap U_k}$$

by (I, Ex. 1.22), so the isomorphisms $\varphi_i^{-1} \circ \varphi_j$ give an element of $\check{H}^1(\mathfrak{U}, \mathcal{O}_X^*)$. Define a map

$$L : \varinjlim_{\mathfrak{U}} \check{H}^1(\mathfrak{U}, \mathcal{O}_X^*) \to \operatorname{Pic} X$$

as follows: for every $\alpha \in \check{H}^1(\mathfrak{U}, \mathcal{O}_X^*)$, define $L(\alpha)$ to be the invertible sheaf on X obtained by glueing $\mathcal{O}_{U_i}$ via the transition functions defined by $\varphi_{ij}(1) = \alpha_{ij}$. This is an isomorphism by the prevoius remarks, hence Pic $X \cong H^1(X, \mathcal{O}_X^*)$ by (Ex. 4.4). □